Chapter 1 Test: Data Types & Feature Understanding

Level 1 Warmup + Level 2 Interviewer Question Bank

Answering Way: Descriptive, Coding, and Scenario-Based answers below are written the way a thoughtful student would explain them out loud - conversational tone, brief code where useful, and always touching on the 'why' behind the concept, not just the mechanics.

Level 1: Warmup Test (10 Questions)

Section A - Multiple Choice Questions (5)

Q1. Which type of variable has categories with a meaningful order, but the gaps between categories are not necessarily equal?

A. Nominal Categorical
B. Ordinal Categorical
C. Discrete Numerical
D. Continuous Numerical

Answer & Explanation

Q2. In pandas, the 'object' dtype most commonly represents:

A. Only numeric values that failed to parse
B. Text strings, which may be free text OR categorical labels
C. Boolean True/False values exclusively
D. Datetime values that haven't been parsed yet

Answer & Explanation

Q3. A numerical column has only 4 unique values across 10,000 rows. What should you suspect?

A. It's definitely a continuous numerical column
B. It's likely a disguised categorical column
C. It must be a datetime column
D. It's an error and should be dropped immediately

Answer & Explanation

Q4. Which of the following is an example of unstructured data?

A. A SQL table with named columns and fixed types
B. A CSV file with rows and columns
C. A folder of customer product review text and photos
D. A JSON object with consistent, well-defined keys

Answer & Explanation

Q5. A categorical column has 500 unique values out of 1,000 rows. What's generally the best encoding approach?

A. One-hot encoding (create 500 new columns)
B. Target encoding, frequency encoding, or hashing
C. Treat it as free text and ignore it
D. Convert it to a boolean column

Answer & Explanation

Section B - Descriptive Questions (3)

Q6. Why is it a mistake to treat an ID column (like customer_id = 1001, 1002...) as a regular numerical feature? Walk through what could actually go wrong.

Answer

Q7. Explain the difference between structured, semi-structured, and unstructured data, with an example of each.

Answer

Q8. When you calculate both the mean and median of a numerical column and they're very different, what does that tell you, and why does it matter for something like missing value imputation?

Answer

Section C - Coding Questions (2)

Q9. Given a DataFrame `df` with a mix of numeric, text, and datetime columns, write code to automatically split the column names into three lists: numerical_cols, categorical_cols, and datetime_cols.

Answer

Q10. Write code that scans a DataFrame's numeric columns and flags any that have fewer than 15 unique values, since these are likely disguised categorical columns.

Answer

Level 2: Interviewer Test (10 Intermediate / Semi-Advanced Questions)

Section A - Scenario-Based MCQ (3)

Q11. *[Scenario]* Scenario: You load a dataset and find a 'zip_code' column stored as int64, with 500 unique values across 1,000 rows. What's the most appropriate next step?

A. Leave it as int64 and feed it directly into a regression model

B. Cast it to string/object, then consider hashing or target encoding since it's medium/high cardinality

C. One-hot encode all 500 values immediately

D. Drop it - any column with more than 100 unique values is useless

Answer & Explanation

Q12. *[Scenario]* Scenario: You compute mean = 4.2 lakhs and median = 1.1 lakhs for a bank's loan amounts, and the top 5% of borrowers are large corporate accounts. Which statistic should drive your reporting and imputation decisions for the retail segment?

A. Mean, because it uses all the data

B. Median, since it isn't distorted by the small number of very large corporate loans

C. Mode, since it's the simplest to compute

D. Standard deviation, since it captures spread

Answer & Explanation

Q13. *[Scenario]* Scenario: An API returns records like {'order_id': 101, 'customer': {'name': 'Amit', 'city': 'Pune'}}. Which pandas function correctly flattens this into a table with columns like 'customer.name' and 'customer.city'?

A. pd.merge()

B. pd.json_normalize()

C. pd.pivot_table()

D. pd.concat()

Answer & Explanation

Section B - Scenario-Based Descriptive (2)

Q14. *[Scenario]* Scenario: Your team included 'machine_id' (an integer like 1001, 1002...) directly as a numerical feature in a regression model predicting failure rate. The model shows a strong negative relationship: higher machine_id → lower failure rate. A colleague is excited about this 'insight'. What's actually going on, and what would you do?

Answer

Q15. *[Scenario]* Scenario: You've just inherited a 140-column dataset from a data partner with zero documentation. Walk through the process you'd follow before doing any modeling, and explain why skipping this step is risky.

Answer

Section C - Coding Questions (3)

Q16. Write a data quality check function that flags rows with an invalid 'age' (outside 0-120) and an unexpected 'gender' value (anything other than 'M' or 'F').

Answer

Q17. Write a function that takes a DataFrame and classifies every column's cardinality into Binary, Low, Medium, High, or Quasi-Unique.

Answer

Q18. Write code to clean a 'salary' column that contains values like '₹50,000', '₹1,20,000', and 'N/A', converting it into a proper float column.

Answer

Section D - Database Scenario-Based Coding (2)

Q19. *[Database Scenario]* Database Scenario: You suspect a 'customers' SQL table has duplicate customer records based on email address. Write a SQL query to find every email that appears more than once, along with how many times it appears.

Answer

Q20. *[Database Scenario]* Database Scenario: Before pulling a 'patients' table into pandas, you want to check the SQL table directly for schema violations - specifically, rows where 'age' is negative or over 120, since this happened once before at the source system. Write a SQL query to surface these rows.